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INVERSE DESIGNED POLARIZATION ROTATOR AND BEAM SPLITTER

Abstract

A polarization rotating and beam splitting photonic device includes a planar waveguide having an input port and output ports disposed in or on a multi-layer semiconductor stack and a polarization rotating and beam splitting components integrated into the planar waveguide. The polarization rotating component includes a first irregular pattern of at least two materials having different refractive indexes. The first irregular pattern is shaped to rotate at least a portion of an optical signal received via the input port from a transverse magnetic (TM) polarization to a transverse electric (TE) polarization. The beam splitting component includes a second irregular pattern shaped to split the optical signal between the output ports. The first and second irregularly shaped patterns are optically coupled and collectively shaped to receive input TE and TM signals multiplexed on the optical signal at the input port and generate output TE signals demultiplexed on the output ports.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to photonic devices, and in particular but not exclusively, relates to polarization rotators and beam splitters.

BACKGROUND INFORMATION

[0002] Artificial intelligence (AI) and machine learning (ML) applications are expected to place high demands on the data bandwidth of future XPUs (e.g., central processing units, graphic processing units, tensor processing units, etc.). In fact, data bandwidth is expected to be the bottleneck for future XPU development. In particular, board-to-board and chip-to-chip interconnects will need to support ever increasing bandwidths. Optical interconnects promise to satisfy this increasing bandwidth need. However, despite the high bandwidth provided by optical interconnects, conventional designs suffer from low data bandwidth density (i.e., data bandwidth per unit area). To improve the data bandwidth density of optical interconnects, photonic integrated circuits need to be reduced in physical size.

[0003] A polarization beam splitter (PBS) is a fundamental building block for high-speed optical interconnects as they enable polarization multiplexing. A PBS is an optical filter that splits an incident beam into two separate beams of different polarizations. In the ideal scenario, these separate beams are fully polarized with orthogonal polarizations. In the context of guided light (e.g., optic fibers), the incident light may include transverse electric (TE) and transverse magnetic (TM) polarizations and in the context of single mode waveguides (e.g., single mode optic fibers), the light may include just the fundamental spatial modes TE**0** and TM**0** for the respective polarizations. The TE**0** and TM**0** signals can increase the bandwidth of guided light by encoding distinct data channels on the orthogonal polarization modes TE**0** and TM**0**.

[0004] A polarization rotator and beam splitter (PRBS) is yet another important building block for high-speed optical interconnects. A PRBS operates to split the TE and TM polarization components of light received on a single physical port to separate physical ports while also converting the disparate polarization components to a common polarization (e.g., TE polarization). PRBS may be integrated into high bandwidth photonic transceivers that use the TE and TM polarizations as independent data channels.

[0005] Conventional PBS have physical sizes on the order of 100 um×8 um while PRBS are typically quite long ranging between 700 um to 1500 um in length. A PBS that is able to substantially reduce these physical dimensions while maintaining expected functional characteristics (e.g., polarization crosstalk & isolation, insertion/transmission loss, back reflection, etc.) will help satisfy the higher data bandwidth density demands expected from future XPU development. Similarly, a PRBS that is capable of achieving a desired performance profile while reducing its physical length can reduce the overall size of a pluggable optical sub-assembly (OSA) into which the PRBS is integrated. The size of these photonic devices is expect to be a limiting factor in future optical communication and computing platforms.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Non-limiting and non-exhaustive embodiments of the invention are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified. Not all instances of an element are necessarily labeled so as not to clutter the drawings where appropriate. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles being described.

[0007] FIG. **1**A is a functional block diagram of a polarization rotator and beam splitter (PRBS), in accordance with an embodiment of the disclosure.

[0008] FIG. **1**B illustrates a practical application of a PRBS in the receiver and transmitter components of a semiconductor photonic transceiver, in accordance with an embodiment of the disclosure.

[0009] FIG. **2**A illustrates cross-sections of different types of planar waveguides into which a PRBS may be integrated, in accordance with embodiments of the disclosure.

[0010] FIG. **2**B is a cross-sectional view of a PRBS that includes an inversed designed irregularly shaped pattern integrated into a planar waveguide, in accordance with an embodiment of the disclosure.

[0011] FIG. **2**C is a cross-sectional view of a PRBS that includes an inversed designed irregularly shaped pattern integrated into a planar waveguide, in accordance with an embodiment of the disclosure.

[0012] FIG. **3**A illustrates a PRBS having polarization rotating and beam splitting components integrated together in a common component with a single combined inversed designed pattern sharing a common overlapping area within a planar waveguide, in accordance with an embodiment of the disclosure.

[0013] FIG. **3**B illustrates a first example irregular pattern capable of implementing both polarization rotating and beam splitting functions of a PRBS, in accordance with an embodiment of the disclosure.

[0014] FIG. **3**C illustrates a second example irregular pattern capable of implementing both polarization rotating and beam splitting functions of a PRBS, in accordance with an embodiment of the disclosure.

[0015] FIG. **3**D illustrates the multipath interferometry that guides and converts an input TM signal to an output TE signal on a first output port of the PRBS, in accordance with an embodiment of the disclosure.

[0016] FIG. **3**E illustrates the asymmetrical power splitting that guides an input TE signal to an output TE signal on a second output port of the PRBS, in accordance with an embodiment of the disclosure.

[0017] FIG. **3**F illustrates a third example irregular pattern capable of implementing both polarization rotating and beam splitting functions of a PRBS, in accordance with an embodiment of the disclosure.

[0018] FIG. **3**G illustrates power density heat maps for the third example showing the TM to TE signal path and the TE to TE path, in accordance with an embodiment of the disclosure.

[0019] FIG. **4**A illustrates a PRBS having distinct polarization beam splitting (PBS) and polarization rotating (PR) components both of which are integrated into a planar waveguide, in accordance with an embodiment of the disclosure.

[0020] FIG. **4**B illustrates an irregularly shaped pattern for implementing a PBS component of a PRBS, in accordance with an embodiment of the disclosure.

[0021] FIGS. **4**C and **4**D illustrate a transverse electric (TE) path through the PBS component that directs a majority of the power of the input TE signal to an output port via asymmetrical power splitting, in accordance with an embodiment of the disclosure.

[0022] FIGS. **4**E and **4**F illustrate transverse magnetic (TM) paths through the PBS component that direct a majority of the power of a TM optical signal to an output port via multipath interferometry, in accordance with an embodiment of the disclosure.

[0023] FIG. **4**G illustrates an irregularly shaped pattern for implementing a PR component of a PRBS, in accordance with an embodiment of the disclosure.

[0024] FIG. **5**A illustrates a PRBS having distinct polarization rotating and beam splitting components that also leverage spatial mode conversions, in accordance with an embodiment of the disclosure.

[0025] FIG. **5**B illustrates examples of fundamental and higher order spatial modes.

[0026] FIG. **6**A illustrates an irregularly shaped pattern for implementing a mode converter that selectively rotates a TM**0** signal to a TE**1** signal, in accordance with an embodiment of the disclosure.

[0027] FIG. **6**B illustrates an irregularly shaped pattern for implementing a mode splitter that splits spatial modes while also converting a higher order mode to a fundamental order mode, in accordance with an embodiment of the disclosure.

[0028] FIG. **7**A illustrates a demonstrative simulated environment for simulating the operation of a PRBS under design, in accordance with an embodiment of the disclosure.

[0029] FIG. **7**B illustrates an operational simulation of a PRBS, in accordance with an embodiment of the disclosure.

[0030] FIG. **7**C illustrates an adjoint simulation (backpropagation) of a performance loss error through the simulated environment including the PRBS, in accordance with an embodiment of the disclosure.
[0031] FIG. **8**A is a flow chart illustrating example time steps for operational and adjoint simulations used to inverse design a PRBS, in accordance with an embodiment of the disclosure.
[0032] FIG. **8**B is a flow chart illustrating a relationship between an operational simulation and the adjoint simulation (backpropagation), in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION

[0033] Embodiments of a system, apparatus, and method of operation for a polarization rotating and beam splitting (PRBS) photonic device are described herein. In the following description numerous specific details are set forth to provide a thorough understanding of the embodiments. One skilled in the relevant art will recognize, however, that the techniques described herein can be practiced without one or more of the specific details, or with other methods, components, materials, etc. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring certain aspects.
[0034] Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.
[0035] A typical function for a PRBS is to split light including both transverse electric (TE) and transverse magnetic (TM) components, which is received on a common physical port, into two separate physical ports while also converting the TM polarization component into a TE polarization component. Conversion of the TM components into TE components is often desirable since photonic devices may operate differently on the different polarizations (e.g., due to birefringence or other polarization dependent photoelectric effects).
[0036] FIG. **1**A illustrates a PRBS **100** in accordance with an embodiment of the disclosure. The illustrated embodiment of PRBS **100** includes a PRBS region **105**, an input port **110**, and output ports **115**A and **115**B (collectively **115**). During operation, an optical signal **101** including TE components and TM components are received from a waveguide **120** (e.g., optic fiber) at input port **110**. In the illustrated embodiment the TE and TM components are propagating in the fundamental spatial modes TE**0** and TM**0**, which may be referred to as input TE and input TM signals, respectively. The input TE and TM signals may be independent (e.g., orthogonal) data channels multiplexed on a common carrier wavelength. Of course, waveguide **120** may guide multiple carrier wavelengths (referred to as dense wavelength division multiplexing or DWDM) each having independent TE and TM channels.
[0037] The PRBS region **105** operates to direct the input TE signal to the physical output port **115**B for launching into waveguide **125**B while both directing the input TM signal to output port **115**A and converting its TM polarization to a TE polarization. In various embodiments described herein, PRBS region **105** is an irregularly shaped pattern of at least two materials having different refractive indexes integrated into a planar waveguide. The planar waveguide itself may be disposed in or on a multi-layer semiconductor stack and even be an integrated component of a photonic integrated circuit (PIC). For example, PRBS **100** may be implemented in a silicon PIC. In one embodiment, the irregularly shaped pattern is an inverse designed pattern. In some embodiments, the irregularly shaped pattern is a single component having a single combined pattern that implements both the polarization rotation and beam splitting functions. In yet other embodiments, the irregularly shaped pattern is separated into two distinct components each having distinct patterns that separately implement the polarization rotation and beam splitting functions. The irregularly shaped patterns may each be implemented as a single layer two-dimensional pattern (illustrated) or implemented as a three-dimensional pattern that extends across and is defined by multi-layers of a semiconductor stack up. Both single-layer and multi-layer patterns may be optimized using inverse design techniques such as those described herein.
[0038] PRBS are important components of silicon photonic optical transceiver modules that enable

polarization multiplexing across optical links. FIG. **1**B illustrates an example optical receiver **130** and optical transmitter **131** which each include an instance of PRBS **100**. Optical receiver **130** and optical transmitter **131** may be integrated into a silicon photonic optical transceiver. As illustrated, optical receiver **130** includes an edge coupler **140** that receives, amplifies, and/or filters an input optical signal including data signals multiplexed on the TE and TM polarization. PRBS **100** demultiplexes the TE and TM polarizations to respective physical ports while also rotating the TM polarization to the TE polarization. The DWDM signals are then demultiplexed by DEMUXs **145** and converted from the optical to electrical realm by photodetectors **150**.

[0039] Optical transmitter **131** operates in a reverse manner to optical receiver **130**. In particular, electrical data signals are modulated by modulators **160**, which in turn are multiplexed onto a common waveguide by MUXs **165**. These WDMD optical signals are multiplexed onto the TE polarization and TE**0** spatial mode. Each MUX **165** outputs its WDMD optical signal to a respective input port of PRBS **100**. It is noteworthy that the principle of reciprocity in electromagnetism means PRBS **100** may be operated in reverse to multiplex TE**0** optical signals received on distinct physical ports onto a single physical port using both the TE and TM polarizations. Finally, edge coupler **170** amplifies and/or filters the TE/TM multiplexed optical signal before launching the optical signal onto a transmission medium. Of course, edge couplers **140** and **170** may further implement various communication protocols related to the transmission medium.

[0040] PRBS **100** may be integrated into a planar waveguide. FIG. **2**A illustrates cross-sections of different types of planar waveguides, including a rib waveguide **205**, a ridge waveguide **210**, a slab waveguide **215**, and a buried channel waveguide **220**, in which PRBS **100** may be disposed. For example, PRBS **100** may be disposed within rib portion **206** of rib waveguide **205**, ridge portion **211** of ridge waveguide **210**, slab portion **216** of slab waveguide **215** or channel portion **221** of buried channel waveguide **220**. FIGS. **2**B and **2**C illustrate specific examples corresponding to ridge waveguide **210** and buried channel waveguide **220**.

[0041] Referring to FIG. **2**B, PRBS **100** may be integrated into rib portion **231** of a rib waveguide **230** disposed within a multi-layer semiconductor stack **235**. FIG. **2**B is a cross-sectional illustration of multi-layer semiconductor stack **235**. The cross-sectional view of FIG. **2**B is aligned with cross-section A-A′ through PRBS region **105** in FIG. **1**A. In one embodiment, multi-layer semiconductor stack **235** is a semiconductor-on-insulator stack including layers of silicon and silicon oxide while PRBS region **105** disposed within rib portion **231** is an irregular pattern of high and low refractive material (e.g., silicon and silicon oxide) fabricated into rib portion **231**. Of course, stack **235** may include other materials and in FIG. **2**B includes a material layer **240** disposed along the top side of rib waveguide **230**. In one embodiment, material layer **240** is a passivation layer (e.g., 12 nm thick) of silicon nitride disposed approximately 5 nm above rib portion **231**. In another embodiment, material layer **240** is polysilicon. Material layer **240** extends in the X-Y plane along at least a portion of PRBS region **105** that is responsible for rotation of the input TM signal to the TE polarization. Material layer **240** is a distinct material having a different refractive index than its surrounding material in multi-layer semiconductor stack **235** and helps encourage polarization rotation by breaking the symmetry of multi-layer semiconductor stack **235** in the vicinity of the planar waveguide (e.g., rib waveguide **230**). In one exemplary embodiment, dimension D**1** is 106 nm while dimension D**2** is 55 nm. Of course, other dimensions are possible as well.

[0042] Referring to FIG. **2**C, PRBS **100** may be integrated into buried channel waveguide **250** disposed within a multi-layer semiconductor stack **255**. FIG. **2**C is a cross-sectional illustration of multi-layer semiconductor stack **255**. The cross-sectional view of FIG. **2**C is aligned with cross-section A-A′ through PRBS region **105** in FIG. **1**A. In one embodiment, multi-layer semiconductor stack **255** is a semiconductor stack including layers of silicon and silicon oxide while PRBS region **105** disposed within buried channel waveguide **250** is an irregular pattern of high and low refractive material (e.g., silicon and silicon oxide) fabricated into buried channel waveguide **250**. Of course, stack **255** may include other materials and in FIG. **2**C includes a material layer **260** disposed offset from buried channel waveguide **250**. In one embodiment, material layer **260** is a distinct material from

the material layers surrounding it or even distinct from the materials that form PRBS region **105**. For example, in one embodiment, material layer **260** is silicon nitride. In another embodiment, material layer **260** is polysilicon. Material layer **260** extends in the X axis along at least a portion of PRBS region **105** that is responsible for rotation of the input TM signal to the TE polarization. Material layer **260** is a distinct material having a different refractive index than its surrounding material in multi-layer semiconductor stack **255** and helps encourage polarization rotation by breaking the symmetry of multi-layer semiconductor stack **255** in the vicinity of the planar waveguide (e.g., buried channel waveguide **250**). In one embodiment, material layer **260** is disposed along just one side of buried channel waveguide **250** with its center **270** offset from the center **275** of buried channel waveguide **250**. In one exemplary embodiment, dimension D**3** is approximately 161 nm while dimension D**4** is approximately 300 nm, and dimension D**5** is 5 nm. Of course, other dimensions are possible as well.
[0043] FIG. **3**A is a functional block diagram illustrating a PRBS **300** having polarization rotating and beam splitting components integrated together in a common component, in accordance with an embodiment. PRBS **300** is one possible implementation of PRBS **100** illustrated in FIG. **1**A. FIGS. **3**B, **3**C, and **3**F illustrate specific implementations of inverse designed irregularly shaped patterns that form PRBS region **305** of PRBS **300**. PRBS **300** achieves the polarization rotating and beam splitting functions with a single combined inverse designed pattern sharing a common overlapping area within any of the planar waveguides illustrated in FIGS. **2**A-C.
[0044] In the illustrated embodiments, PRBS region **305** is a planar waveguide region having an irregularly shaped pattern disposed within the planar waveguide as a two-dimensional (2D) pattern. Of course, in other embodiments, a three-dimensional (3D) pattern may also be implemented where a multi-layer pattern is optimized via inverse design. In the illustrated embodiment, the 2D pattern is defined using two materials having distinct refractive indexes (e.g., silicon and silicon oxide). The pattern is an irregular pattern. For example, at the macro-level, the irregular pattern is not formed by regular geometric shapes such as triangles, rectangles, pentagons, hexagons, octagons, etc. Rather, the pattern is an organic pattern that resembles channels, inlets, and islands of a natural coastline. Of course, at the micro-level, the pattern may be formed by pixelated deposits of the two or more materials, which individual pixels may comprise a geometric shape. The feature size and shape of the individual material pixels is dependent upon the fabrication process, but the overall pattern does not resemble a simple, regular geometric shape such as a triangle, rectangle, or other low order polygon (e.g., 10 sides or less).
[0045] In one embodiment, PRBS **300** is inspired by inverse design. In particular, the pattern of at least two materials having differing refractive indexes may be defined by an iterative minimization of a loss function that sums a transmission loss, a reflection loss, and a crosstalk loss. The optimization objective of the inverse design methodology may be constructed as a function of the following loss function Loss(x),
[00001]
$$\mathrm{Loss}(x) = \text{.Math.}\ \mathrm{Transmissionloss}(x,\ ) + \text{.Math.}\ \mathrm{Reflectionloss}(x,\ ) + \text{.Math.}\ \mathrm{Crosstalkloss}(x,\ ),$$
$$\text{where, } \mathrm{Transmissionloss}(x,\ ) = (\mathrm{Transmission}(x,\ ) - \mathrm{targetvalues1})^2$$
$$\mathrm{Reflectionloss}(x,\ ) = (\mathrm{Reflection}(x,\ ) - \mathrm{targetvalues2})^2$$
$$\mathrm{Crosstalkloss}(x,\ ) = (\mathrm{Crosstalk}(x,\ ) - \mathrm{targetvalues3})^2\ .$$
The objective is constructed in a way that the resulting structure/pattern of PRBS region **305** is encouraged to guide TE optical signals received at physical input port **110** to physical output port **115**B while guiding TM optical signals received at physical input port **110** to physical output port **115**A and rotate the TM polarization to a TE polarization.
[0046] FIG. **3**A assigns virtual port (VP) labels to the TE and TM polarizations at each physical port. For example, input port **110** is assigned virtual ports VP**1** and VP**2** for the TE and TM polarizations, respectively. Similarly, virtual ports VP**3** and VP**4** are assigned to output port **115**B for the TE and TM polarizations, respectively, and virtual ports VP**5** and VP**6** are assigned to output port **115**A for the TE and TM polarizations, respectively. FIG. **3**A further illustrates a scattering matrix (s-matrix) **310** having columns and rows corresponding to the virtual port labels. The behavior of PRBS region **305**

may be described using the scattering-parameters (s-parameters) populated within s-matrix **310**. S-matrix **310** may be referenced to populate the loss function Loss(x) used to design the PRBS region **305** using inverse design techniques. In particular, the s-parameters within s-matrix **310** represent the target values for transmission (T), reflection (R), and crosstalk (C). Blank spaces in the s-matrix **310** indicate “don't care” values that are not used as a target value in the loss function Loss(x).

[0047] Inverse design operates using a design simulator (aka design model) configured with an initial design or pattern for PRBS region **305** to perform a forward operational simulation of the initial design (e.g., using Maxwell's equations for electromagnetics). For example, the initial design could be a random pattern of silicon and silicon dioxide. The output of the forward operational simulation is a simulated field response at output ports **115** in response to stimuli at input port **110**. Specific performance parameters of this output field response may be selected as parameters of interest (e.g., power loss, wavelength, crosstalk, etc.) and are referred to as simulated performance parameters. The simulated performance parameters are used by the loss function to calculate a performance loss value, which may be a scalar value (e.g., mean square difference between simulated performance values and target values). The differentiable nature of the design model enables a backpropagation via an adjoint simulation of a performance loss error, which is the difference between the simulated output values and the desired/target performance values. The performance loss error is backpropagated through the design model during the adjoint simulation to generate structural gradients that represent, for example, the sensitivity of the performance loss value to changes in the structural material properties (e.g., topology or pattern of materials) of PRBS region **305**. A program such as TensorFlow published by Google may be used to calculate the gradients. These gradients may then be used by a structural optimizer to optimize or refine the initial structural design to generate a revised structural design of PRBS region **305**. The forward and reverse simulations may then be iterated along with the structural optimization (e.g., iterative gradient descent, stochastic gradient descent, etc.) until the performance loss value falls within acceptable design criteria (referred to as saturation) and/or for a predetermined number of iterations. The above description is merely an example inverse design technique that may be used to refine or optimize the features and topology of the pattern within PRBS region **305**. It is appreciated that other inverse design techniques alone, or in combination with other conventional design techniques, may also be implemented.

[0048] The inverse design techniques described above may be applied to determine the specific material combinations, feature sizes, and feature arrangement (i.e., irregular pattern) to achieve the desired polarization rotation and beam splitting using the above loss function. Loss(x) is a function of x, where x is a vector representing the structural pattern of materials having different refractive indexes within PRBS region **305**.

[0049] FIG. **3**B illustrates a first example irregular pattern **315** capable of implementing both polarization rotating and beam splitting functions of a PRBS, in accordance with an embodiment. Irregular pattern **315** is one possible implementation of PRBS region **305** (and **105**). As mentioned, irregular pattern **315** is a single combined pattern that is integrated into a single common area of a planar waveguide. Irregular pattern **315** is an inverse designed pattern shaped to demultiplex the TE and TM signals received in input port **110** by directing a power majority of the TE**0** signal to output port **115**B via asymmetrical power splitting while directing a power majority of the TM**0** signal to output port **115**A via multipath interferometry. The inverse designed irregular pattern **315** may be fabricated from a patterned combination of silicon (being a higher refractive index material represented in white) and silicon oxide (being a lower refractive index material represented in black).

[0050] FIG. **3**C illustrates another example inverse designed irregular pattern **320** that performs the same demultiplexing of the input TE and TM signals as described in connection with FIG. **3**B. Irregular pattern **320** is yet another possible implementation of PRBS region **305** (and **105**). FIG. **3**D is a power density heat map illustrating the power density of the TM**0** to TE**0** multipath interferometry between input port **110** and output port **115**A for irregular pattern **320**. Similarly, FIG. **3**E illustrates the power density of the TE**0** to TE**0** asymmetric power splitting path between input port **110** and output port **115**B for irregular pattern **320**. In one embodiment, irregular pattern **320** has approximate dimensions of D**1**=30 um and D**2**=7 um. Of course, other dimensions may be implemented as well. For example, D**1** may range between 5-300 um and D**2** may range between 2-40 um.

[0051] FIG. **3**F illustrates yet another example inverse designed irregular pattern **350** that performs the same demultiplexing of the input TE and TM signals as described in connection with FIG. **3**B while also rotating the TM polarization to the TE polarization. Irregular pattern **350** is yet another possible implementation of PRBS region **305** (and **105**). Irregular pattern **350** includes a number of irregularly shaped features defined by the refractive materials. Many of these features may deviate from the illustrated pattern while achieving nearly similar efficient operation. However, some notable features include an irregularly shaped channel **355** of higher index material (e.g., silicon) extending continuously and circuitously from input port **110** to output port **115**B. In contrast, it is also notable that the TM**0** to TE**0** path from input port **110** to output port **115**A does not include a continuous uninterrupted channel of the higher index material extending between the two ports. Rather, output port **115**A is connected to a small fjord-like feature **357** of the higher index material. Pattern **350** also includes a plurality of other irregularly shaped "islands" of the higher index material disposed within and throughout the lower index material (e.g., silicon oxide). These irregularly shaped islands are separate and distinct from irregularly shaped channel **355**. The irregularly shaped islands are more densely dispersed proximate to the distal ends of irregular pattern **350** close to the ports while being more sparse in the middle region between the input and output ports. Pattern **350** is non-symmetrical along both orthogonal axes. In one embodiment, irregular pattern **350** has approximate dimensions of D**11**=30 um and D**12**=8 um. Of course, other dimensions may be implemented as well.

[0052] FIG. **3**G includes a power density heat map **360** illustrating approximate power density of the TM**0** to TE**0** multipath interferometry between input port **110** and output port **115**A for irregular pattern **350**. FIG. **3**G also illustrates a power density heat map **370** for the TE**0** to TE**0** asymmetric power splitting path between input port **110** and output port **115**B for irregular pattern **350**. As can be seen, a strong majority of the optical power in the input TM**0** signal is directed towards the region coincident with output port **115**A while a strong majority of the optical power in the input TE**0** signal is directed towards the region coincident with output port **115**B.

[0053] FIG. **4**A is a functional block diagram illustrating a PRBS **400** having distinct polarization beam splitting (PBS) and polarization rotating (PR) components linked end-to-end, in accordance with an embodiment. PRBS **400** is one possible implementation of PRBS **100** illustrated in FIG. **1**A. PRBS **400** includes a PBS **405** and PR **410** that collectively implement PRBS region **105** illustrated in FIG. **1**A by linking PBS **405** and PR **410** end-to-end at an intermediate port **415**. In one embodiment, PBS **405** and PR **410** are implemented by distinct irregularly shaped patterns that are both integrated end-to-end into a single common planar waveguide (or a pair of linked planar waveguides) by optically linking a TM**0** output of PBS **405** to the input of PR **410**. FIGS. **4**B-**4**F illustrate a specific inverse designed irregularly shaped pattern that forms PBS **405** while FIG. **4**G illustrates a specific inverse designed irregularly shaped pattern that forms PR **410**.

[0054] Collectively, PBS **405** and PR **410** achieve the beam splitting and polarization rotation functions in two distinct patterns/components. PBS **405** beam splits the input TE and TM signals to output port **115**B and intermediate port **415**, respectively. PBS **405** directs a first power majority of the input TE signal to output port **115**B via asymmetrical power splitting while directing a second power majority of the input TM signal to intermediate port **415** via multipath interferometry. PR **410** includes an input aligned with intermediate port **415** and its output is aligned to output port **115**A. PR **410** operates to rotate the TM polarization to a TE polarization.

[0055] FIG. **4**B illustrates an example inversely designed irregularly shaped pattern **420**, in accordance with an embodiment of the disclosure. Pattern **420** is one possible inverse design implementation for PBS **405** illustrated in FIG. **4**A. Pattern **420** is an irregular pattern of at least two materials (e.g., silicon and silicon dioxide) having different refractive indexes. Pattern **420** is shaped to demultiplex the TE optical signals (e.g., TE**0**) and the TM optical signals (e.g., TM**0**) received at input port **110**. Pattern **420** is shaped to direct a majority of the power of the TE optical signal receive at input port **110** to output port **115**B. The selective directing of the TE optical signal to output port **115**B is achieved via an asymmetrical power splitting that sends a majority of the TE power to output port **115**B. Correspondingly, pattern **420** is also shaped to direct a majority of the power of the TM optical signal received at input port **110** to intermediate port **415**. The selective directing of the TM optical signal to intermediate port **415** is achieved via multipath interferometry that sends a majority of the TM power to intermediate port **415**.

[0056] In one embodiment, pattern **420** is integrated into a planar waveguide having the illustrated pattern disposed within the planar waveguide as a two-dimensional (2D) pattern. Of course, in other embodiments, a three-dimensional (3D) pattern defined within multi-layers of a semiconductor stack up may also be implemented. In the illustrated embodiment, the 2D pattern is defined using two materials having distinct refractive indexes (e.g., silicon and silicon dioxide). Pattern **420** is an irregular pattern. For example, at the macro-level, the irregular pattern is not formed by regular geometric shapes such as triangles, rectangles, pentagons, hexagons, octagons, etc. Rather, pattern **420** is an organic pattern that resembles channels, inlets, and islands of a natural coastline. Of course, at the micro-level, the pattern may be formed by pixelated deposits of the two or more materials, which individual pixels may comprise a geometric shape. The feature size and shape of the individual material pixels is dependent upon the fabrication process, but the overall pattern does not resemble a simple, regular geometric shape such as a triangle, rectangle, or other low order polygon (e.g., 10 sides or less). In one embodiment, pattern **420** has approximate dimensions of D**3**=7 um and D**4**=8 um. Of course, other dimensions may be implemented as well.

[0057] In the illustrated embodiment, the irregular shaped pattern includes a number of pattern features (i.e., irregularly shaped features) that facilitate the asymmetrical power splitting of the TE optical signal along with the simultaneous multipath interferometry of the TM optical signal. FIGS. **4**C, **4**D, **4**E, and **4**F illustrate these pattern features. In particular, FIGS. **4**C and **4**D illustrate a TE path **425** through polarization splitting region **430** that directs a majority of the power of the TE optical signal to output port **115**B via asymmetrical power splitting. Polarization splitting region **430** is the multi material region within a planar waveguide into which pattern **420** is integrated. Correspondingly, FIGS. **4**E and **4**F illustrate TM paths **440**A-C through polarization splitting region **430** that direct a majority of the power of the TM optical signal to intermediate port **415** via multipath interferometry.

[0058] Referring to FIG. **4**D, TE path **425** extends from input port **110** to output port **115**B along an irregular shaped channel **450**. Irregular shaped channel **450** is formed of the higher index material, such as silicon (illustrated as white), surrounded by the lower index material, such as silicon dioxide (illustrated as black). In the illustrated embodiment, irregular shaped channel **450** includes an S-bend shape that is not obstructed between input port **110** and output port **115**B by the lower index material. In other words, the irregular shaped channel **450** forms a continuous, unobstructed, pathway that forms a mild circuitous S-bend shape between the ports. Irregular shaped channel **450** facilitates the asymmetrical power splitting that guides a majority of the optical power in the TE optical signal from input port **110** to output port **115**B. FIG. **4**C is a power density heat map illustrating the power density of the TE optical signal (in-plane E-field) through polarization splitting region **430**. As illustrated, a strong majority of the optical power in the TE optical signal is guided along irregular shaped channel **450** along TE path **425** and reaches output port **115**B.

[0059] Referring to FIGS. **4**E and **4**F, TM paths **440**A-C (collectively referred to as TM paths **440**) extend from input port **110** to intermediate port **415** along multiple interferometry paths. TM paths **440**A-C are also defined by the higher index material, such as silicon (illustrated as white), surrounded by the lower index material, such as silicon dioxide (illustrated as black). The primary TM path **440**A is composed of two sub-paths that weave around and through a number of scattering locations. In other words, primary TM path **440**A includes a plurality of scattering location **447** formed from islands of lower index material disposed within the high index material. The secondary TM path **440**B takes an unobstructed circuitous path through the higher index material. The tertiary TM path **440**C takes an even less direct path through the higher index material (partially along irregular shaped channel **450**) and is obstructed at location **460** by low index material. It is believed that evanescent coupling and/or the near normal incidence of tertiary TM path **440**C to the low index material obstruction permits the components of the TM optical signal to pass through the lower index material obstruction at location **460**. Regardless, the multiple TM paths established by pattern **420** between the input port **110** and intermediate port **415** direct a strong majority of the optical power in just the TM optical signal from input port **110** to intermediate port **415** via multipath interferometry. The power density heat map of FIG. **4**E illustrates the strong coupling of the TM optical signal (out-of-plane E-field) to intermediate port **415**.

[0060] FIG. **4**G illustrates an irregularly shaped pattern **470** for implementing PR **410**, in accordance with an embodiment. Pattern **470** is an inverse design irregularly shaped pattern that operates to rotate a TM signal received on intermediate port **415** to a TE signal output from output port **115**A. Accordingly, PR **410** is coupled end-to-end with PBS **405** by having its input and output optical aligned with intermediate port **415** and output port **115**A, respectively, to form PRBS **400**. Pattern **470** includes a higher refractive index material (illustrated as white) surrounded by a lower refractive index material (illustrated as black). In the illustrated embodiment, pattern **470** does not include a continues uninterrupted channel of the higher refractive index material extending between its input and output ports. Pattern **470** is non-symmetrical along both orthogonal axes and includes irregularly shaped features defined by the refractive materials. In one embodiment, pattern **470** has approximate dimensions of D**5**=20 um and D**6**=7 um. Of course, other dimensions may be implemented as well. For example, D**5** may range between 5-300 um and D**6** may range between 2-40 um. Power density heat map **475** illustrates the internal power density while rotating the TM signal to the output TE signal.

[0061] FIG. **4**G assigns virtual port (VP) labels to the TE and TM polarizations at each physical port. For example, intermediate port **415** is assigned virtual ports VP**1** and VP**2** for the TE and TM polarizations, respectively. Similarly, virtual ports VP**3** and VP**4** are assigned to output port **115**A for the TE and TM polarizations, respectively. S-matrix **480** has columns and rows corresponding to these virtual port labels. The behavior of pattern **470** may be described using the s-parameters populated within s-matrix **480**. S-matrix **480** may be referenced to populate the loss function Loss(x) (see above) used to design the pattern **470** using inverse design techniques. In particular, the s-parameters within s-matrix **480** represent the target values for transmission (T), reflection (R), and crosstalk (C). Blank spaces in the s-matrix **480** indicate “don't care” values that are not used as a target value in the loss function Loss(x).

[0062] FIG. **5**A is a functional block diagram illustrating a PRBS **500** having distinct polarization rotating and beam splitting components linked end-to-end, in accordance with an embodiment. In addition to the polarization rotating and beam splitting functions, PRBS **500** achieves the overall PRBS function by further incorporating a mode conversion into each of its distinct components. Accordingly, the polarization rotating and mode conversion component is referred to as mode converter **505** while the beam splitting and mode conversion component is referred to as mode splitter **510**.

[0063] Mode conversion is the conversion of optical power propagating in one spatial mode within a waveguide to another spatial mode. FIG. **5**B illustrates spatial modes, including a fundamental spatial mode **520** propagating within a planar waveguide **525** and higher order spatial modes **530** or **535** that may also be propagating within planar waveguide **525**. The fundamental spatial mode having a TE polarization is referred to as the TE**0** mode and the fundamental spatial mode having a TM polarization is referred to as the TM**0** mode. Correspondingly, the higher order spatial modes of the TE polarization are referred to as the TE**1** mode, TE**2** mode, etc. while the higher order spatial modes of the TM polarization are referred to as the TM**1** mode, TM**2** mode, etc.

[0064] Returning to FIG. **5**A, PRBS **500** is one possible implementation of PRBS **100** illustrated in FIG. **1**A. PRBS **500** includes a mode converter **505** and mode splitter **510** that collectively implement PRBS region **105** illustrated in FIG. **1**A by linking mode converter **505** and mode splitter **510** end-to-end at an intermediate port **515**. In one embodiment, mode converter **505** and mode splitter **510** are implemented by distinct irregularly shaped patterns that are both integrated end-to-end into a single planar waveguide (or a pair of end-to-end connected planar waveguides) by optically linking the output of mode converter **505** to the input of mode splitter **510**. FIG. **6**A illustrates an example inverse designed irregularly shaped pattern for implementing mode converter **505** while FIG. **6**B illustrates an example inverse designed irregularly shaped pattern that implements mode splitter **510**.

[0065] Collectively, mode converter **505** and mode splitter **510** achieve the beam splitting and polarization rotation functions in two distinct patterns/components. Mode converter **505** passes the input TE**0** mode to its output at intermediate port **515** without changing its polarization or spatial mode. In contrast, mode converter **505** rotates the polarization and converts the spatial mode of the input TM**0** signal. In particular, mode converter **505** rotates the polarization of the input TM**0** signal to the TE polarization and upconverts its spatial mode to the TE**1** mode.

[0066] Mode splitter **510** operates as a beam splitter that splits the optical signal received at intermediate port **515** according to spatial mode. In particular, optical power propagating in the TE**0** fundamental spatial mode at intermediate port **515** is directed to output port **115**B while optical power propagating in the TE**1** higher order spatial mode is directed to output port **115**A. However, mode splitter **515** further operates as a mode converter for the higher order spatial modes of the TE polarization. Accordingly, in addition to beam splitting, mode splitter **510** converts the optical power propagating in the higher order spatial mode TE**1** received on intermediate port **515** down to the TE**0** fundamental spatial mode at output port TE**0**.

[0067] FIG. **5**A assigns VP labels to the TE and TM polarizations at each physical port for mode converter **505** and mode splitter **510**. For mode converter **505**, input port **110** is assigned virtual ports VP**1** and VP**2** for the TE and TM polarization, respectively, and intermediate port **515** is assigned virtual ports VP**3** and VP**4** for the TE**0** and TE**1** spatial modes, respectively. Similarly, for mode splitter **510**, intermediate port **515** is assigned virtual ports VP**1** and VP**2** for spatial modes TE**0** and TE**1**, respectively, output port **115**B is assigned virtual ports VP**3** and VP**4** for the TE**0** and TE**1** spatial modes, respectively, and output port **115**A is assigned virtual ports VP**5** and VP**6** for the TE**0** and TE**1** spatial modes, respectively. S-matrix **540** has columns and rows corresponding to the virtual port labels for mode converter **505** while s-matrix **550** has columns and rows corresponding to the virtual port labels for mode splitter **510**. The behavior of the inverse designed pattern of mode converter **505** may be described using the s-parameters populated within s-matrix **540** while the behavior of the inverse designed pattern of mode splitter **510** may be described using the s-parameter populated within s-matrix **550**. S-matrices **540** and **550** may be referenced to populate the loss function Loss(x) (see above) used to design the patterns of mode converter **505** and mode splitter **510**. In particular, the s-parameters represent the target values for transmission (T), reflection (R), and crosstalk (C). Blank spaces in s-matrices **540** and **550** indicate “don't care” values that are not used as a target value in the loss function Loss(x).

[0068] FIG. **6**A illustrates an irregularly shaped pattern **600** for implementing a mode converter that selectively rotates a TM**0** signal to the TE**1** mode, in accordance with an embodiment of the disclosure. Pattern **600** represents one possible implementation of mode converter **505**. Pattern **600** is an inverse designed irregularly shaped pattern of at least two materials having different refractive indexes (e.g., silicon and silicon oxide). During operation, the input TE**0** signal received at input port **110** is passed to intermediate port **515** without changing the polarization or spatial mode of the input TE**0** signal. However, pattern **600** is shaped to selectively rotate the polarization of a TM**0** signal received at input port **110** to the TE polarization at intermediate port **515** while also converting its spatial mode to the higher order TE**1** mode. Pattern **600** is non-symmetrical along both orthogonal axes and includes irregularly shaped features defined by the refractive materials. In one embodiment, pattern **600** has approximate dimensions of D**7**=7 um and D**8**=20 um. Of course, other dimensions may be implemented as well. For example, D**8** may range between 5-300 um and D**7** may range between 2-40 um.

[0069] FIG. **6**B illustrates an irregularly shaped pattern **601** for implementing a mode splitter that splits a spatial mode to different output ports while also converting the higher order spatial mode to a fundamental spatial mode, in accordance with an embodiment of the disclosure. Pattern **601** represents one possible implementation of mode splitter **510**. During operation, pattern **601** is shaped to split the

TE**0** signal received at intermediate port **515** from the TE**1** signal and pass the TE**0** signal to output port **115**B via channel **605** without changing the polarization or spatial mode of the TE**0** signal. Correspondingly, pattern **601** is further shaped to split the TE**1** signal from the TE**0** signal, direct the TE**1** signal to the output port **115**A, and convert its spatial mode from the higher order spatial mode to the fundamental spatial mode TE**0**. In one embodiment, pattern **601** has approximate dimensions of D**9**=5 um and D**10**=30 um. Of course, other dimensions may be implemented as well. For example, D**9** may range between 5-300 um and D**10** may range between 5-300 um.

[0070] Pattern **601** is an inverse designed irregularly shaped pattern of at least two materials having different refractive indexes. In one embodiment, these materials include a higher index material (e.g., silicon) and a lower index material (e.g., silicon oxide) arranged into the illustrated pattern. Pattern **601** may be fabricated using conventional semiconductor deposition and etching techniques. Pattern **601** is non-symmetrical along both orthogonal axes and includes irregularly shaped features defined by the refractive materials. The white portions represent the higher refractive index material while the black portions represent the lower refractive index material. Although pattern **601** may assume a variety of different dimensions, in one embodiment, D**9** is approximately 7 um and D**10** is approximately 24 um.

[0071] Pattern **601** includes a number of irregularly shaped features defined by the refractive materials. Many of these features may deviate from the illustrated pattern while achieving nearly similar efficient operation. However, some notable features include an irregularly shaped channel **605** of higher index material extending continuously and circuitously from intermediate port **515** to output port **115**B. In contrast, it is also notable that the TE**1** to TE**0** path from intermediate port **515** to output port **115**A does not include a continuous uninterrupted channel of the higher index material extending between the two ports. Rather, output port **115**A is connected to a fjord-like feature **610** of the higher index material. Pattern **601** also includes a plurality of other irregularly shaped “islands” of the higher index material disposed within and throughout the lower index material. These irregularly shaped islands are separate and distinct from irregularly shaped channel **605**.

[0072] The inverse design techniques described above may be applied to determine the specific material combinations, feature sizes, and feature arrangement (i.e., pattern) to achieve the desired polarization rotation, beam splitting, and mode conversion using the above loss function Loss(x). Loss(x) is a function of x, where x is a vector representing the structural pattern of materials having different refractive indexes within the design region.

[0073] FIGS. **7**A-**7**C illustrate an initial setup, an operational simulation, and an adjoint simulation of a simulated environment **701**, respectively, for optimizing structural parameters of a physical device (e.g., PRBS **100**, PRBS region **305**, pattern **315**, pattern **320**, PBS **405**, PR **410**, pattern **420**, pattern **470**, mode converter **505**, mode splitter **510**, pattern **600**, or pattern **601**) with a design model, in accordance with an inverse design embodiment. The simulated environment **701** and corresponding initial setup, operational simulation, adjoint simulation, and structural parameter optimization may be achieved via a physics simulator using Maxwell's equations. As illustrated in FIGS. **7**A-**7**C, the simulated environment is represented in two-dimensions, however it is appreciated that higher dimensionality (e.g., 3-dimensional space) may also be used to describe the simulated environment **701** and the physical device. In some embodiments, the optimization of the structural parameters of the physical device illustrated in FIGS. **7**A-**7**C may be achieved via, inter alia, simulations (e.g., time-forward and backpropagation) that utilize a finite-difference time-domain (FDTD) method to model the field responses (e.g., both electric and magnetic).

[0074] FIG. **7**A illustrates an example rendering of a simulated environment **701**-A describing an electromagnetic device. The simulated environment **701**-A represents the simulated environment **701** at an initial time step (e.g., an initial set up) for optimizing structural parameters of the physical device. The physical device described by the simulated environment **701** may correspond to any of the PRBS, PBS, PR, mode converter, or mode splitter described above having a designable region **705** (e.g., irregular shaped pattern) in which the structural parameters of the simulated environment may be designed, modified, or otherwise changed. The simulated environment **701** includes an excitation source **715** (e.g., a gaussian pulse, a wave, a waveguide mode response, and the like) at the location of input port **110**. The electrical and magnetic fields (e.g., field response) within the simulated environment **701** may change in response to the excitation source **715**. The specific settings of the initial structural parameters, excitation source, performance parameters, and other metrics (i.e., initial description) for a first-principles simulation of a physical device are input before the operational simulation starts.

[0075] As illustrated, the simulated environment **701** (and subsequently the physical device under design) is described by a plurality of voxels **710**, which represent individual elements of the two-dimensional (or three-dimensional) space of the simulated environment. Each of the voxels is illustrated as two-dimensional squares, however it is appreciated that the voxels may be represented as cubes or other shapes in three-dimensional space. It is appreciated that the specific shape and dimensionality of the plurality of voxels **710** may be adjusted dependent on the simulated environment **701**. It is further noted that only a portion of the plurality of voxels **710** are illustrated to avoid obscuring other aspects of the simulated environment **701**. Each of the plurality of voxels **710** is associated with one or more structural parameters, a field value to describe a field response, and a source value to describe the excitation source at a specific position within the simulated environment **701**. The field response, for example, may correspond to a vector describing the electric and/or magnetic field at a particular time step for each of the plurality of voxels **710**. More specifically, the vector may correspond to a Yee lattice that discretizes Maxwell's equations for determining the field response. In some embodiments, the field response is based, at least in part, on the structural parameters and the excitation source **715**.

[0076] FIG. **7**B illustrates an example operational simulation of the simulated environment **701**-B at a particular time step in which the excitation source **715** is active (e.g., generating waves originating at the excitation source **715** that propagate through the simulated environment **701**). As mentioned, the physical device is an optical modulator operating at the frequency of interest and having a particular waveguide mode (e.g., transverse electromagnetic mode, transverse electric mode, etc.) and the excitation source is at an input port **110**. The operational simulation occurs over a plurality of time steps. When performing the operational simulation, changes to the field response (e.g., the field value) for each of the plurality of voxels **710** are updated in response to the excitation source **715** and based, at least in part, on the structural parameters of the physical device at each of the plurality of time steps. Similarly, in some embodiments the source value is updated for each of the plurality of voxels (e.g., in response to the electromagnetic waves from the excitation source **715** propagating through the simulated environment). It is appreciated that the operational simulation is incremental and that the field value (and source value) is updated incrementally at each time step as time moves forward for each of the plurality of time steps. It is further noted that in some embodiments, the update is an iterative process and that the update of each field and source value is based, at least in part, on the previous update of each field and source value.

[0077] When performing the operational simulation, the performance loss function, Loss(x), may be computed at each output port **720** and **725** based, at least in part, on a comparison (e.g., mean squared difference) between the field response and a desired field response at a designated time step (e.g. a final time step of the operational simulation). A performance loss value may be described in terms of a specific performance value (e.g., power). Structural parameters may be optimized for this specific performance value.

[0078] FIG. **7**C illustrates an example backpropagation of performance loss error backwards within the simulated environment **701**-C describing the physical device. In one embodiment, the adjoint performance simulation injects a performance loss error at output ports **720** and **725** as a sort of reverse excitation source for stimulating a reverse field response through voxels **710** of simulated environment **701**-C. The adjoint performance simulation of the performance loss error determines an influence that changes in the structural parameters of voxels **710** have on the performance loss value.

[0079] FIG. **8**A is a flow chart **800** illustrating example time steps for a time-forward simulation **810** and backpropagation **850** within a simulated environment, in accordance with an embodiment of the present disclosure. Flow chart **800** is one possible implementation that a design model may use to perform a forward operational simulation **810** and backpropagation **850** of a simulated environment. In the illustrated embodiment, the forward operational simulation utilizes a FDTD method to model the field response (both electric and magnetic) at a plurality of time steps in response to an excitation source. More specifically, the time-dependent Maxwell's equations (in partial differential form) are discretized to solve for field vector components (e.g. the field response of each of the plurality of voxels **710** of the simulated environment **701** in FIGS. **7**A-**7**C) over a plurality of time steps.

[0080] As illustrated in FIG. **8**A, the flow chart **800** includes update operations for a portion of operational simulation **810** and adjoint simulation **850**. Operational simulation **810** occurs over a plurality of time-steps (e.g., from an initial time step to a final time step over a pre-determined or conditional number of time steps having a specified time step size) and models changes (e.g., from the initial field values **811**) in electric and magnetic fields of a plurality of voxels describing the simulated environment and/or physical device that collectively correspond to the field response. More specifically, update operations (e.g., **812**, **814**, and **816**) are iterative and based on the field response, structural parameters **804**, and one or more physical stimuli sources **808**. Each update operation is succeeded by another update operation, which are representative of successive steps forward in time within the plurality of time steps. For example, update operation **814** updates the field values **813** (see, e.g., FIG. **7**B) based on the field response determined from the previous update operation **812**, sources **808**, and the structural parameters **804**. Similarly, update operation **816** updates the field values (see, e.g., FIG. **8**B) based on the field response determined from update operation **814**. In other words, at each time step of the operational simulation the field values (and thus field response) are updated based on the previous field response and structural parameters of the physical device. Once the final time step of the operational simulation **810** is performed, the loss value **818** may be determined (e.g., based on a pre-determined loss function **820**). The loss gradients determined from block **852** may be treated as adjoint or virtual sources (e.g., physical stimuli or excitation source originating at an output region) which are backpropagated in reverse (from the final time step incrementally through the plurality of time steps until reaching the initial time step) to determine structural gradient **868**.

[0081] In the illustrated embodiment, the FDTD solve (e.g., time-forward simulation **810**) and backpropagation **850** problem are described pictorially, from a high-level, using only “update” and “loss” operations as well as their corresponding gradient operations. The simulation is set up initially in which the structure parameters, the excitation source, and the initial field states of the simulated environment (and electromagnetic device) are provided. As discussed previously, the field states are updated in response to the excitation source based on the structural parameters. More specifically, the update operation is given by ϕ, where custom-character=ϕ(custom-character,custom-character, custom-character) for custom-character=1, . . . custom-character. Here, custom-character corresponds to the total number of time steps (e.g., the plurality of time steps) for the time-forward simulation, custom-character corresponds to the field response (the field value associated with the electric and magnetic fields of each of the plurality of voxels) of the simulated environment at time step custom-character, custom-character corresponds to the excitation source(s) (the source value associated with the electric and magnetic fields for each of the plurality of voxels) of the simulated environment at me step custom-character, and custom-character corresponds to the structural parameters describing the topology and/or material properties of the electromagnetic device.

[0082] It is noted that using the FDTD method, the update operation can specifically be stated as:

[00002] $$(x_i, b_i, z) = A(z)x_i + B(z)b_i. \quad (1)$$

That is to say the FDTD update is linear with respect to the field and source terms. Concretely, A(custom-character)∈custom-character.sup.N×N and B(custom-character)∈custom-character.sup.N×N are linear operators which depend on the structure parameters, custom-character and act on the fields, custom-character, and the sources, custom-character, respectively. Here, it is assumed that custom-character, custom-character∈custom-character.sup.N where N is the number of FDTD field components in the time-forward simulation. Additionally, the loss operation is given by L=(custom-character, . . . , custom-character), which takes as input the computed fields and produces a single, real-valued scalar (e.g., the loss value) that can be reduced and/or minimized.

[0083] In terms of revising or otherwise optimizing the structural parameters of the electromagnetic device, the relevant quantity to produce is custom-character, which is used to describe the change in the loss value with respect to a change in the structural parameters of the electromagnetic device and is denoted as the "structural gradient" illustrated in FIG. **8**A.

[0084] FIG. **8**B is a chart **880** illustrating the relationship between the update operation for the operational simulation and the adjoint simulation (e.g., backpropagation), in accordance with an embodiment of the present disclosure. More specifically, FIG. **8**B summarizes the operational and adjoint simulation relationships that are involved in computing the structural gradient, custom-character, which include custom-character,

[00003] $\frac{\partial x_{i+1}}{\partial x_i}$,

custom-character and custom-character. The update operation **814** of the operational simulation updates the field values **813**, custom-character of the plurality of voxels at the custom-characterth time step to the next time step (i.e., custom-character+1 time step), which correspond to the field values **815**, custom-character. The gradients **855** are utilized to determine custom-character for the backpropagation (e.g., update operation **856** backwards in time), which combined with the gradients **869** are used, at least in part, to calculate the structural gradient, custom-character, custom-character is the contribution of each field to the loss value, L. It is noted that this is the partial derivative, and therefore does not take into account the causal relationship of

[00004] $x_i \text{ .fwdarw. } x_{i+1}$.

Thus,

[00005] $\frac{\partial x_{i+1}}{\partial x_i}$

is utilized which encompasses the

[00006] $x_i \text{ .fwdarw. } x_{i+1}$

relationship. The loss gradient, custom-character may also be used to compute the structural gradient, custom-character, and corresponds to the total derivative of the field with respect to loss value, L. The loss gradient, custom-character, at a particular time step, custom-character, is equal to the summation of

[00007] $\frac{\partial L}{\partial x_i} + \frac{dL}{dx_{i+1}}\frac{\partial x_{i+1}}{\partial x_i}$.

Finally, custom-character, corresponds to the field gradient, is used which is the contribution to custom-character from each time/update step. custom-character is given by:

[00008] $\frac{dL}{dz} = \text{.Math.}_i \frac{dL}{dx_i}\frac{\partial x_i}{\partial z}$. (2)

[0085] For completeness, the full form of the first time in the sum, custom-character, is expressed as:

[00009] $\frac{dL}{dx_i} = \frac{\partial L}{\partial x_i} + \frac{dL}{dx_{i+1}}\frac{\partial x_{i+1}}{\partial x_i}$. (3)

Based on the definition of ϕ as described by equation (1), it is noted that

[00010] $\frac{\partial x_{i+1}}{\partial x_i} = A(z)$,

which can be substituted in equation (3) to arrive at an adjoint update for backpropagation (e.g., the update operations such as update operation **856**), which can be expressed as:

[00011] $\frac{dL}{dx_i} = \frac{\partial L}{\partial x_i} + \frac{dL}{dx_{i+1}}A(z)$, (4) or $\nabla_{x_i} L = A(z)^T \nabla_{x_{i+1}} L + \frac{\partial L^T}{\partial x_i}$. (5)

The adjoint update is the backpropagation of the loss gradients from later to earlier time steps and may be referred to as a backwards solve for custom-character. The second term in the sum of the structural gradient, custom-character, is denoted as:

[00012] $\frac{\partial x_i}{\partial z} = \frac{d\ (x_{i-1} \text{ .Math. } b_{i-1}, z)}{dz} = \frac{dA(z)}{dz}x_{i-1} + \frac{dB(z)}{dz}b_{i-1}$, (6) [0086] for the particular form of ϕ described by equation (1).

[0087] The above description of illustrated embodiments of the invention, including what is described in the Abstract, is not intended to be exhaustive or to limit the invention to the precise forms disclosed.

While specific embodiments of, and examples for, the invention are described herein for illustrative purposes, various modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize.
[0088] These modifications can be made to the invention in light of the above detailed description. The terms used in the following claims should not be construed to limit the invention to the specific embodiments disclosed in the specification. Rather, the scope of the invention is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation.

Claims

1. A polarization rotating and beam splitting (PRBS) photonic device, comprising: a planar waveguide disposed in or on a multi-layer semiconductor stack, the planar waveguide having an input port and output ports; a polarization rotating component integrated into the planar waveguide, wherein the polarization rotating component includes a first irregular pattern of at least two materials having different refractive indexes, the first irregular pattern shaped to rotate at least a portion of an optical signal received via the input port from a transverse magnetic (TM) polarization to a transverse electric (TE) polarization; and a beam splitting component integrated into the planar waveguide, wherein the beam splitting component includes a second irregular pattern of the at least two materials shaped to split the optical signal between the output ports, wherein the first and second irregularly shaped patterns are optically coupled and collectively shaped to receive input TE and TM signals multiplexed on the optical signal at the input port of the planar waveguide and generate output TE signals demultiplexed on the output ports.
2. The PRBS photonic device of claim 1, further comprising: a material layer disposed within the multi-layer semiconductor stack and extending along at least a portion of the polarization rotating component to break a symmetry of the multi-layer semiconductor stack in a vicinity of the planar waveguide to encourage polarization rotation between the TM and TE polarizations within the polarization rotating component, wherein the material layer is physically offset from the planar waveguide and is a distinct material from the at least two materials.
3. The PRBS photonic device of claim 2, wherein the at least two materials comprise silicon and silicon dioxide and the material layer comprises silicon nitride or polysilicon.
4. The PRBS of claim 2, wherein the material layer comprises a passivation layer of the multi-layer semiconductor stack or the material layer comprises a buried channel waveguide having a first center that is laterally offset from a second center of the planar waveguide.
5. The PRBS of claim 1, wherein the planar waveguide comprises one of a rib waveguide, a ridge waveguide, a slab waveguide, or a buried channel waveguide.
6. The PRBS of claim 5, wherein the planar waveguide comprises either a rib waveguide or a ridge waveguide and the first and second irregular patterns are disposed within a rib portion of the rib waveguide or a ridge portion of the ridge waveguide.
7. The PRBS of claim 1, wherein the first and second irregular patterns are inverse designed patterns formed from silicon and silicon dioxide disposed within the planar waveguide, wherein the first and second irregular patterns are optically coupled end-to-end within the planar waveguide, and wherein the multi-layer semiconductor stack comprises a photonic integrated circuit (PIC).
8. The PRBS of claim 1, wherein the polarization rotating and beam splitting components are integrated together as a common component and the first and second irregular patterns form a single combined pattern sharing a common overlapping area within the planar waveguide.
9. The PRBS of claim 1, wherein the beam splitting component comprises a polarization beam splitter (PBS) and the second irregular pattern is shaped to demultiplex the input TE and TM signals by directing a first power majority of the input TE signal received at the input port to a first one of the output ports via asymmetrical power splitting while directing a second power majority of the input TM signal received at the input port to an intermediate port between the beam splitter component and

the polarization rotating component via multipath interferometry.

10. The PRBS of claim 9, wherein the second irregular pattern comprises a first irregular shaped channel of a higher refractive index material surrounded by a lower refractive index material, the first irregular shaped channel extending between the input port and the first one of the output ports.

11. The PRBS of claim 9, wherein the second irregularly shaped pattern is shaped to selectively guide the second power majority of the input TM signal from the input port to the intermediate port via a plurality of TM paths extending from the input port to the intermediate port, the TM paths each defined by a higher refractive index material surrounded by a lower refractive index material.

12. The PRBS of claim of claim 9, wherein the polarization rotating component includes a polarization rotating input aligned with the intermediate port and a polarization rotating output aligned with a second one of the output ports, wherein the first irregular pattern includes a higher refractive index material surrounded by a lower refractive index material and does not include a continues uninterrupted channel of the higher refractive index material extending between the polarization rotating input and output.

13. The PRBS of claim 1, wherein the polarization rotating component comprises a mode converter and the first irregular pattern is shaped to both rotate the input TM signal to the TE polarization at an intermediate port disposed between the polarization rotating and beam splitter components and also shaped to convert a fundamental spatial mode of the input TM signal to a higher order TE spatial mode at the intermediate port.

14. The PRBS of claim 1, wherein the beam splitting component comprises a mode splitter and the second irregular pattern is shaped to both split the optical signal received at an intermediate port disposed between the polarization rotating and beam splitting components to the output ports and to convert the portion of the optical signal rotated by the polarization rotating component from a higher order TE spatial mode at the intermediate port to one of the output TE signals in a fundamental TE spatial mode at one of the output ports.

15. The PRBS of claim 14, wherein the second irregular pattern of the mode splitter includes a first irregularly shaped channel of a higher refractive index material surrounded by a lower refractive index material that extends continuously and circuitously from the intermediate port to a first one of the output ports and does not include a continuous channel of the higher index material extending between the intermediate port and a second one of the output ports.

16. The PRBS of claim 1, wherein at least one of the first or second irregular patterns comprises a three-dimensional pattern that extends across and is defined by multiple layers of the multi-layer semiconductor stack.

17. A polarization rotating and beam splitting (PRBS) photonic device, comprising: a planar waveguide disposed in or on a multi-layer semiconductor stack, the planar waveguide having an input port and output ports; an inverse designed irregular pattern of at least two materials having different refractive indexes, the inverse designed irregular pattern shaped to rotate at least a portion of an optical signal received via the input port from a transverse magnetic (TM) polarization to a transverse electric (TE) polarization and further shaped to split the optical signal between the output ports, wherein the inverse designed irregular pattern is further shaped to receive input TE and TM signals multiplexed on the optical signal at the input port of the planar waveguide and generate output TE signals demultiplexed on the output ports.

18. The PRBS of claim 17, wherein the planar waveguide comprises either a rib waveguide or a ridge waveguide and the inverse designed irregular pattern is disposed within a rib portion of the rib waveguide or a ridge portion of the ridge waveguide.

19. The PRBS photonic device of claim 18, further comprising: a material layer disposed within the multi-layer semiconductor stack and extending along at least a portion of the inverse designed irregular pattern to break a symmetry of the multi-layer semiconductor stack in a vicinity of the planar waveguide to encourage polarization rotation between the TM and TE polarizations, wherein the material layer is physically offset from the planar waveguide and is a distinct material from the at least two materials.

20. The PRBS photonic device of claim 19, wherein the at least two materials comprise silicon and silicon dioxide and the material layer comprises silicon nitride.

21. The PRBS photonic device of claim 17, wherein the inversed design irregular pattern is shaped to demultiplex the TE and TM signals by directing a first power majority of the TE signal received at the input port to a first one of the output ports via asymmetrical power splitting while directing a second power majority of the TM signal received at the input port to a second one of the output ports via multipath interferometry.